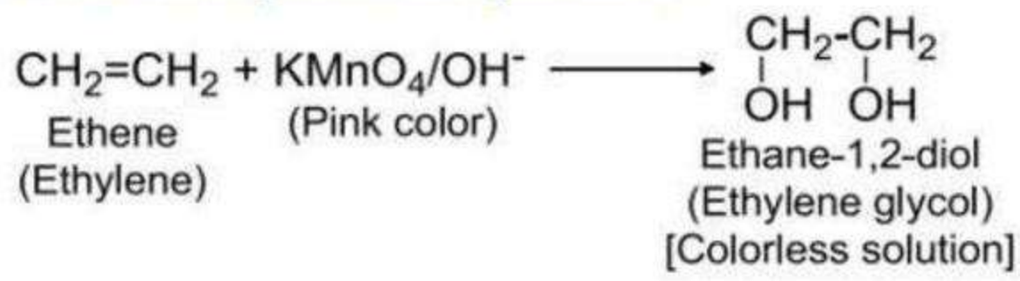
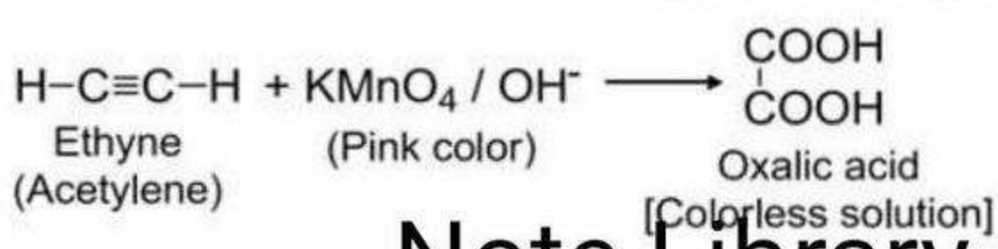
a. Baeyer's reagent test

An alkaline potassium permanganate solution is known as Bayer's reagent. When Bayer's reagent is treated with alkene or alkyne then the pink color of $KMnO_4$ is discharged into colorless solution. This is called unsaturation test as it is used to test the presence of multiple bond in organic compounds.





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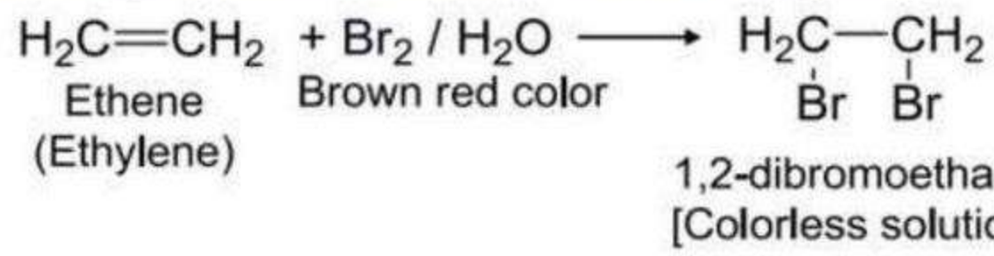
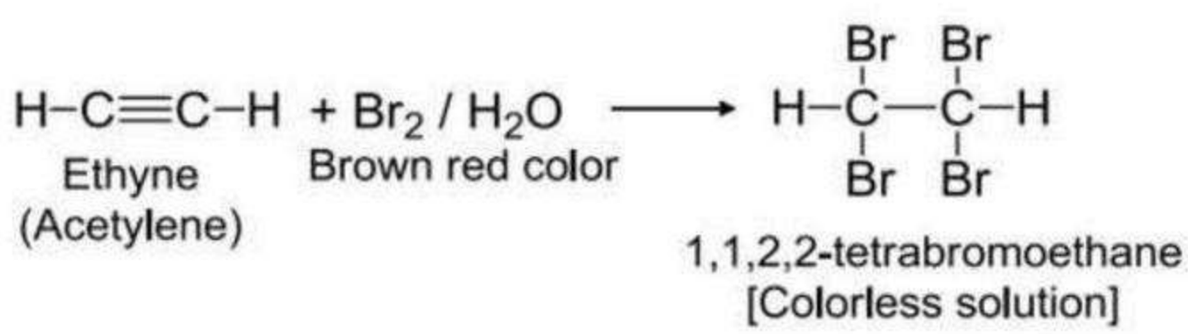
b. Bromine water test

When aqueous bromine or bromine dissolved in CCl_4 solution is treated with alkene and alkyne then brown red color of bromine is discharged to colorless solution. This indicates the presence of multiple bonds in alkenes and alkynes.





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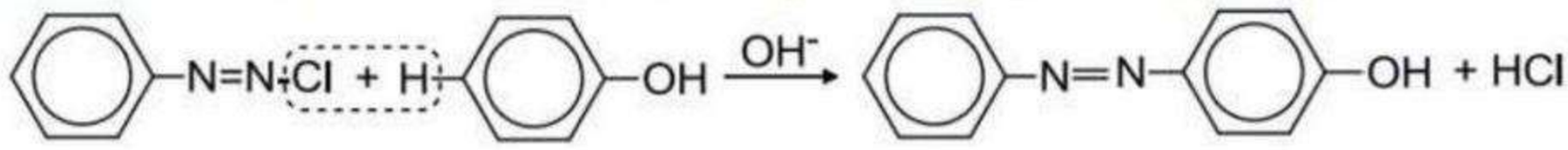
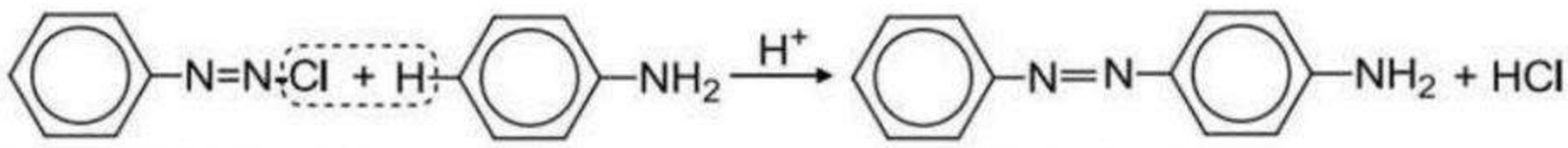
18.Coupling test or reaction for phenol and aniline

When phenol or aniline reacts with benzene diazonium chloride in alkaline or acidic medium then orange-yellow or orange-red crystalline ppt. is formed. This test is also called azo-dye or dye-azo coupling test.



Benzene diazonium chloride | Phenol | p-hydroxyazobenzene (An orange yellow dye)



Benzene diazonium chloride | Aniline | p-aminoazobenzene (An orange red dye)

19.Nitrous acid test for 1°, 2° and 3° nitroalkanes

When primary, secondary and tertiary nitroalkanes are treated with nitrous acid under low temperature followed by boiling with aqueous alkali then three different colours are formed.

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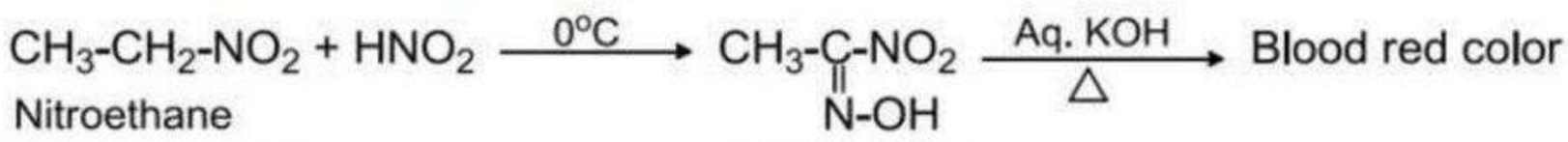
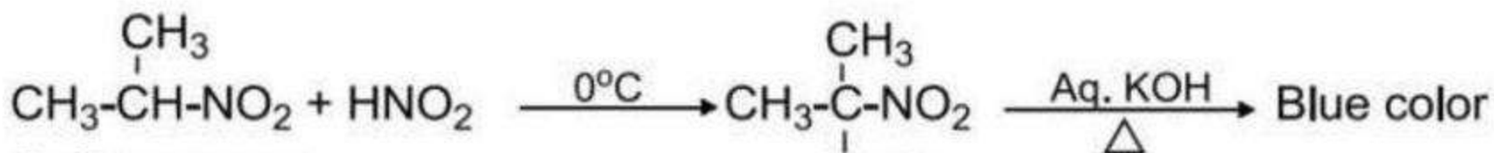
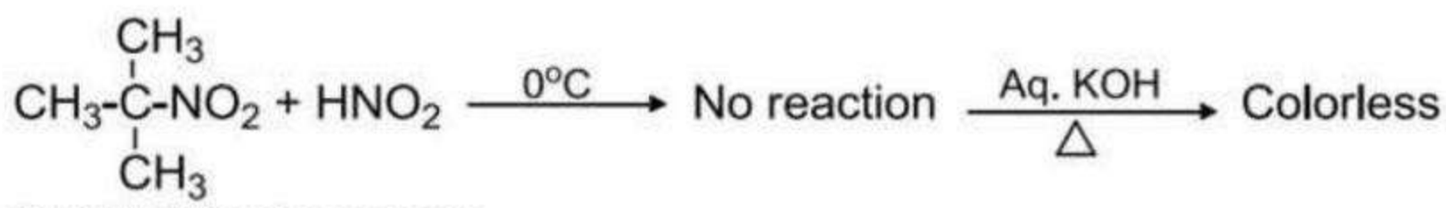
- Primary nitroalkane gives blood red colour.
- Secondary nitroalkane gives blue colour.
- Tertiary nitroalkane gives colourless solution.



Nitroethane (1° nitroalkane) | (Nitrolic acid)



2-nitropropane (2° nitroalkane) | (Pseudonitrol)



2-methyl-2-nitropropane (3° nitroalkane)

20.Unsaturation test for saturated and unsaturated organic compounds

Distinction of saturated and unsaturated organic compounds (Alkanes from alkenes and alkynes) can be done by following two tests.

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$$R\text{-}NH_2 + HNO_2 \longrightarrow R\text{-}OH + H_2O + N_2\uparrow$$

Primary amine → Primary alcohol

$$CH_3CH_2\text{-}NH_2 + HNO_2 \longrightarrow CH_3CH_2\text{-}OH + H_2O + N_2\uparrow$$

Ethyl amine → Ethyl alcohol

$$R_2\text{-}NH + HNO_2 \longrightarrow R_2\text{-}N\text{-}N{=}O + H_2O$$

Secondary amine → N,N-dialkyl nitrosoamine (Water insoluble oily liquid)

$$CH_3\text{-}\underset{\underset{CH_3}{|}}{N}\text{-}H + HNO_2 \longrightarrow CH_3\text{-}\underset{\underset{CH_3}{|}}{N}\text{-}N{=}O + H_2O$$

Dimethyl amine → N,N-dimethyl nitrosoamine

$$R_3N + HNO_2 \longrightarrow R_3NH^+NO_2^-$$

Tertiary amine → Trialkyl ammonium nitrite (Colorless water soluble salt)

$$CH_3\text{-}\underset{\underset{CH_3}{|}}{\ddot{N}}\text{-}CH_3 + HNO_2 \longrightarrow \left[CH_3\text{-}\overset{\overset{H}{\uparrow}}{\underset{\underset{CH_3}{|}}{N}}\text{-}CH_3\right]^+ NO_2^-$$

Trimethyl amine → Trimethyl ammonium nitrite

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17.Distinction of 1°, 2° and 3° amines by Hoffmann's method

When diethyl oxalate reacts with three different classes of amines then three different products are formed.

- Primary amine gives solid dialkyl oxamide
- Secondary amine gives oily liquid dialkyl oxamic ester.
- Tertiary amine does not react with diethyl oxalate.

$$\begin{matrix} C\text{-}OOC_2H_5 \\ | \\ C\text{-}OOC_2H_5 \end{matrix} + 2CH_3\text{-}NH_2 \longrightarrow \begin{matrix} C\text{-}ONH\text{-}CH_3 \\ | \\ C\text{-}ONH\text{-}CH_3 \end{matrix} + 2C_2H_5OH$$

Diethyl oxalate; Methyl amine (1°) → Dimethyl oxamide (Solid product)

$$\begin{matrix} C\text{-}OOC_2H_5 \\ | \\ C\text{-}OOC_2H_5 \end{matrix} + CH_3\text{-}NH\text{-}CH_3 \longrightarrow \begin{matrix} C\text{-}O\text{-}\overset{\overset{CH_3}{|}}{N}\text{-}CH_3 \\ | \\ C\text{-}OOC_2H_5 \end{matrix} + C_2H_5OH$$

Diethyl oxalate; Dimethyl amine (2°) → Dimethyl oxamic ester (Oily Liquid product)

$$\begin{matrix} C\text{-}OOC_2H_5 \\ | \\ C\text{-}OOC_2H_5 \end{matrix} + CH_3\text{-}\overset{\overset{CH_3}{|}}{N}\text{-}CH_3 \longrightarrow \text{No reaction}$$

Diethyl oxalate; Trimethyl amine (3°)

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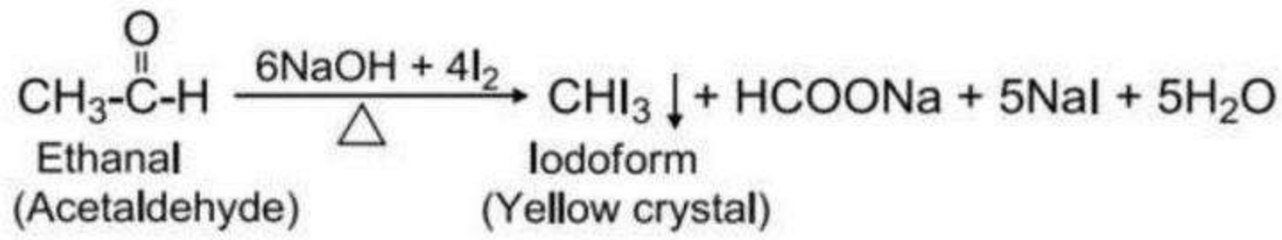
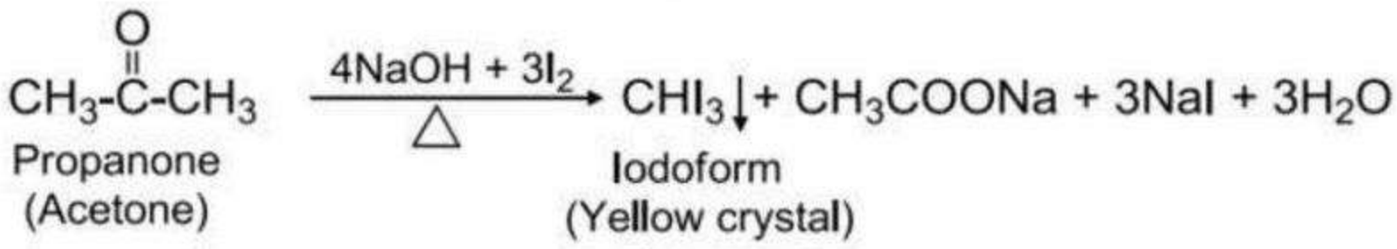
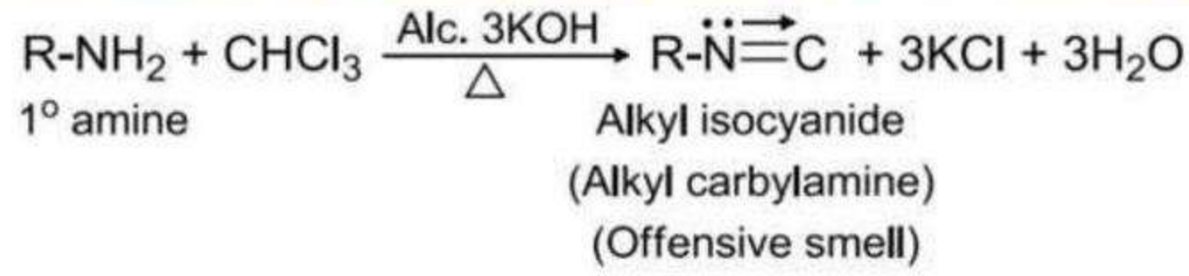
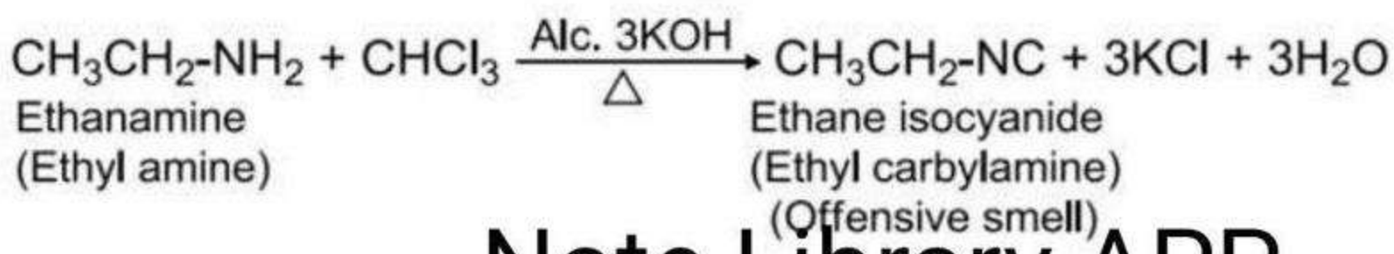




15. Carbylamine test or reaction for primary amine and chloroform (Isocyanide test or reaction)

When primary amine (Aliphatic and aromatic amine) is boiled with chloroform in presence of alcoholic NaOH or KOH then carbylamine having offensive smell is formed. This test is used to identify primary amine in lab i.e, secondary and tertiary amines do not give this test.





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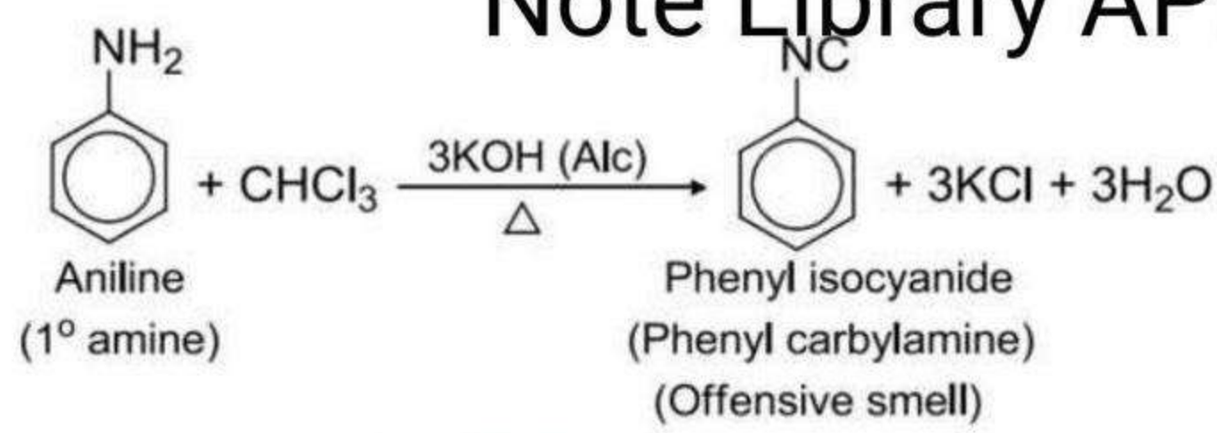

16. Distinction of 1°, 2° & 3° amines by nitrous acid test

When nitrous acid ($NaNO_2$ + HCl at low temperature) react with three different classes of amines then three different products are formed. Therefore by the help of this test, primary, secondary and tertiary amine can be distinguished at laboratory.

- 1° Amines + HNO_2 ⟶ Nitrogen Gas and alcohol
- 2° Amines + HNO_2 ⟶ Water Insoluble Oily liquid
- 3° Amines + HNO_2 ⟶ A Clear water soluble salt

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When aldehyde is heated with Tollen's reagent then the Tollen's reagent is reduced to metallic silver which is deposited on inner wall of test tube. This test or reaction is known as silver mirror test or Tollen's test for aldehyde. This test is largely used to distinguish aldehydes from ketones i.e, aldehyde only gives positive silver mirror test.

$$\underset{\text{Aldehyde}}{R\text{-}\overset{O}{\overset{\|}{C}}\text{-}H} + \underset{\text{Tollen's reagent}}{[Ag(NH_3)_2OH]} \xrightarrow{\Delta} \underset{\text{Silver mirror}}{2Ag\downarrow} + R\text{-}\overset{O}{\overset{\|}{C}}\text{-}ONH_4 + 3NH_3 + H_2O$$

$$\underset{\substack{\text{Ethanal}\\ \text{(Acetaldehyde)}}}{CH_3\text{-}\overset{O}{\overset{\|}{C}}\text{-}H} + [Ag(NH_3)_2OH] \xrightarrow{\Delta} \underset{\text{Silver mirror}}{2Ag\downarrow} + CH_3\text{-}\overset{O}{\overset{\|}{C}}\text{-}ONH_4 + 3NH_3 + H_2O$$

13.Fehling's test or reaction for aldehydes

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An alkaline copper sulphate solution mixed with sodium-potassium tartarate (Rochelle's salt) is called Fehling's solution. When aldehyde is heated with Fehling's solution then brick red ppt. of cuprous oxide (Cu_2O) is formed. This test is used to identify aldehyde and ketone in lab i.e, aldehyde only gives positive Fehling's solution test.

$$\underset{\text{Aldehyde}}{R\text{-}CHO} + \underset{\text{Fehling's solution}}{[2Cu^{++} + 4OH^-]} \xrightarrow{\Delta} \underset{\substack{\text{Cuprous oxide}\\ \text{(Brick red ppt.)}}}{Cu_2O\downarrow} + R\text{-}COOH + 2H_2O$$

$$\underset{\substack{\text{Ethanal}\\ \text{(Acetaldehyde)}}}{CH_3\text{-}\overset{O}{\overset{\|}{C}}\text{-}H} + \underset{\text{Fehling's solution}}{[2Cu^{++} + 4OH^-]} \xrightarrow{\Delta} \underset{\substack{\text{Cuprous oxide}\\ \text{(Brick red ppt.)}}}{Cu_2O\downarrow} + CH_3\text{-}\overset{O}{\overset{\|}{C}}\text{-}OH + 2H_2O$$

14.Iodoform test or reaction for aldehydes and Ketones

When aldehyde and ketone having α-methyl keto group is warmed with iodine in presence of aqueous NaOH then yellow crystal iodoform having hospital smell is formed. This reaction is very useful to detect aldehyde and ketone containing α-methyl keto group.

$H_3C-\overset{O}{\overset{\|}{C}}-$ group
α-methyl keto group

⟸ This group of aldehyde and ketone gives positive iodoform test.

Ethanal (Acetaldehyde) is only one aldehyde that gives positive iodoform test. Ketone having 2-keto group gives iodoform with aqueous NaOH and I_2.

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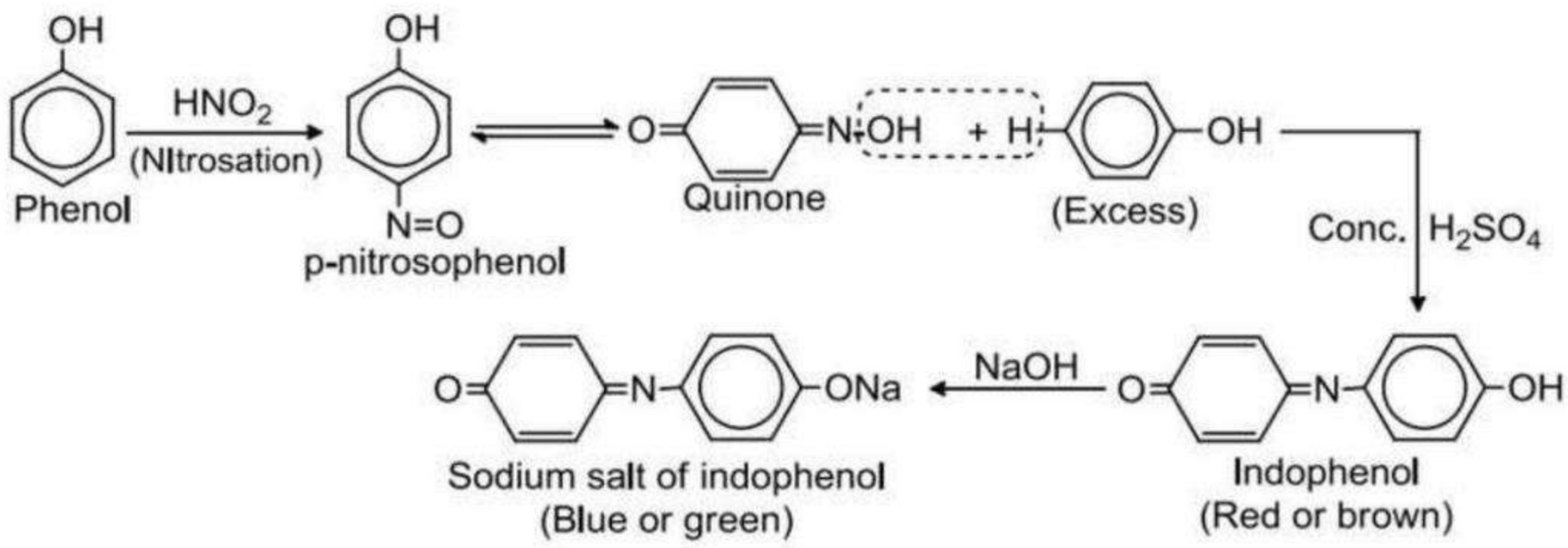
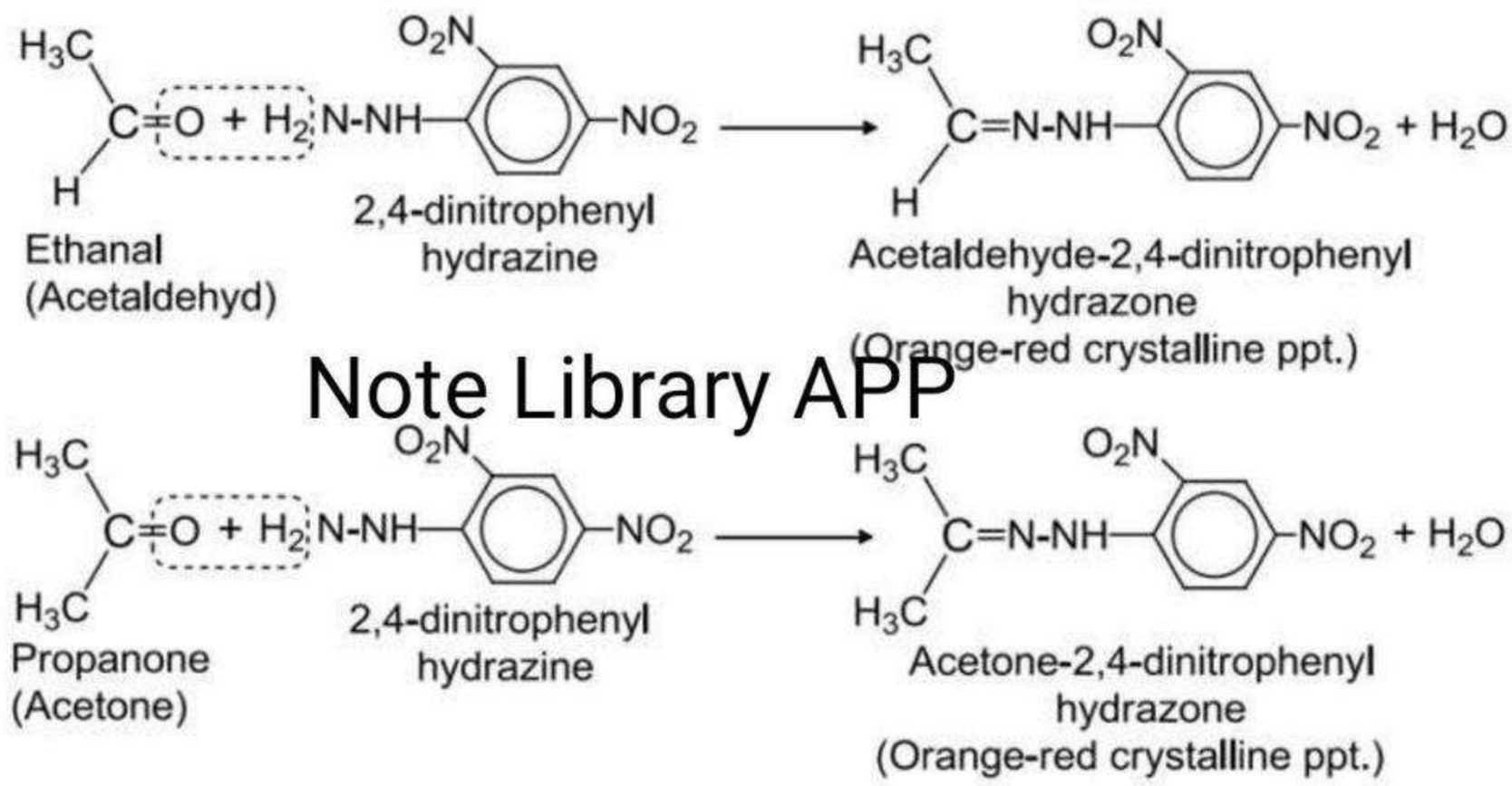


11.2,4-DNP test or reaction for aldehydes and ketones

When aldehyde and ketone reacts with 2,4-dinitrophenylhydrazine (Brady's reagent) then yellow or orange or red colored crystalline ppt. of 2,4-dinitrophenylhydrazone is formed. This test is used to distinguish aldehyde and ketone (carbonyl compounds) from other classes of organic compounds i.e, carbonyl compounds only show positive 2,4-DNP test.



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12.Tollen's test or reaction for aldehydes

An ammonical $AgNO_3$ solution is commonly known as Tollen's reagent.

$$AgNO_3 + NaOH \longrightarrow AgOH\downarrow + NaNO_3$$

Silver hydroxide

$$AgOH + 2NH_4OH \longrightarrow [Ag(NH_3)_2OH] + 2H_2O$$

Diammine silver (I) hydroxide
(Tollen's reagent)

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7. $NaHCO_3$ or Na_2CO_3 test for carboxylic acids

The reaction of Na_2CO_3 and $NaHCO_3$ with carboxylic acid is the test reaction of carboxylic acid as they produce the effervescence of CO_2 gas. Phenol does not produce the effervescence of CO_2 gas with $NaHCO_3$ or Na_2CO_3 solution.

$$CH_3\text{-}COOH + NaHCO_3 \longrightarrow CH_3\text{-}COONa + H_2O + CO_2\uparrow$$

$$2CH_3\text{-}COOH + Na_2CO_3 \longrightarrow 2CH_3\text{-}COONa + H_2O + CO_2\uparrow$$

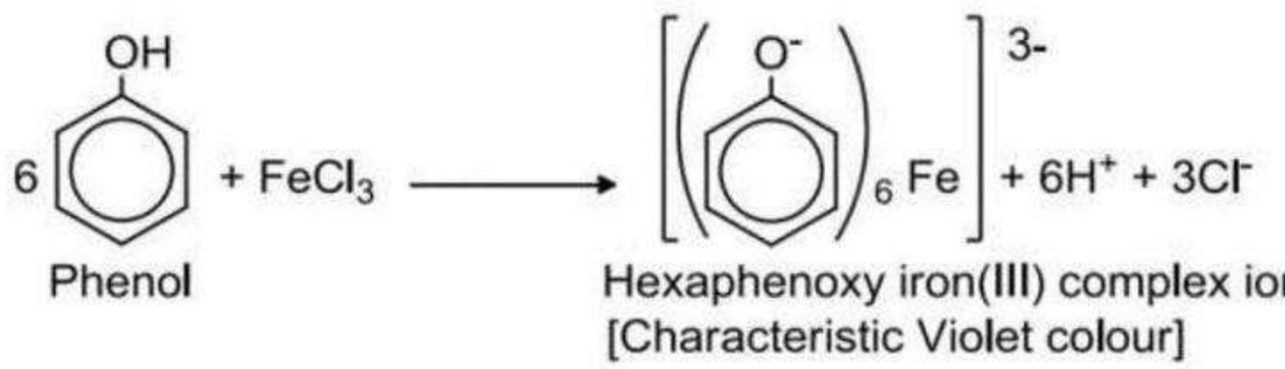
Ethanoic acid (Acetic acid) → Sodium ethanoate (Sodium acetate)

8. $FeCl_3$ test for phenols

When phenol reacts with aqueous $FeCl_3$ solution then characteristic violet colored complex salt is formed. This is the laboratory test of phenol i.e, this test is used to distinguish phenol from other classes of organic compounds.

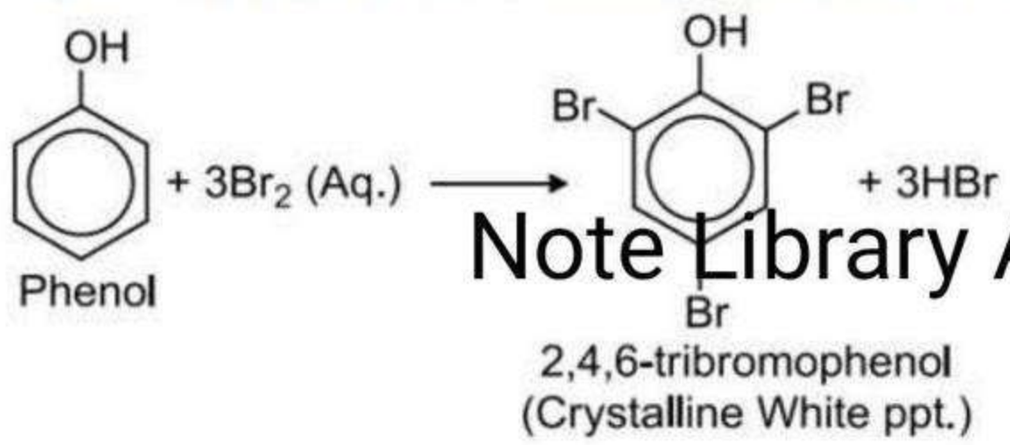
$$6\,C_6H_5OH + FeCl_3 \longrightarrow [(C_6H_5O)_6\text{-}Fe]^{3-} + 6H^+ + 3Cl^-$$

Phenol → Hexaphenoxy iron(III) ion [Violet colour]



9. Bromine water test for phenols

When phenol reacts with bromine water or aqueous bromine then white crystalline ppt. of tri-bromophenol is formed.



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10. Libermann's test for phenols

Phenol gives red brown color on warming with sodium nitrite and conc. H_2SO_4. The red or brown color is changed into intense blue or green on the addition of NaOH, which indicates the presence of phenol.

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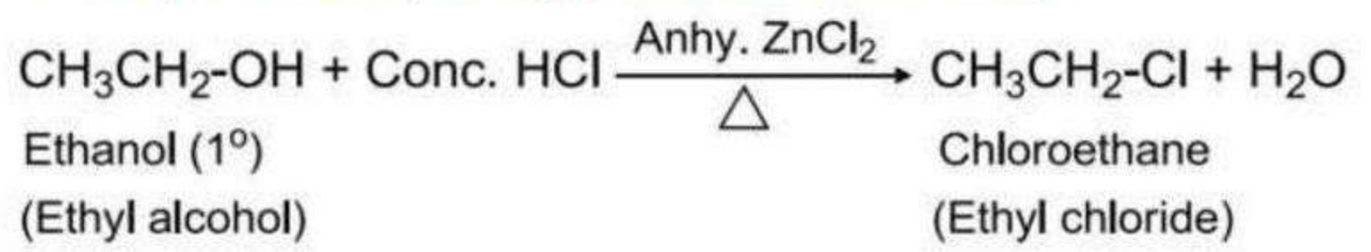
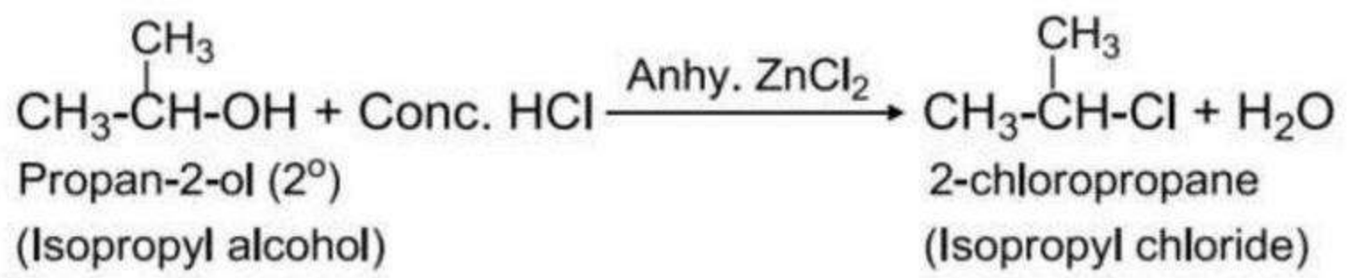
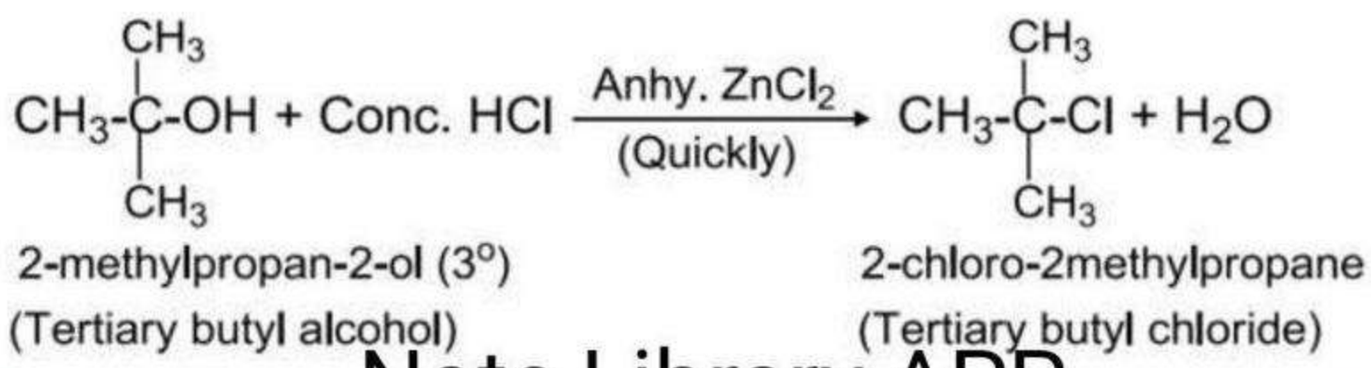
4. Lucas reagent test for alcohols

The mixture of Conc. HCl and anhydrous $ZnCl_2$ is known as Lucas reagent. When Lucas reagent is added to alcohols then three different products are formed.

- Primary alcohol gives white turbidity only after the mixture is heated.
- Secondary alcohol gives white turbidity after 3-5 minutes.
- Tertiary alcohol quickly gives white turbidity.







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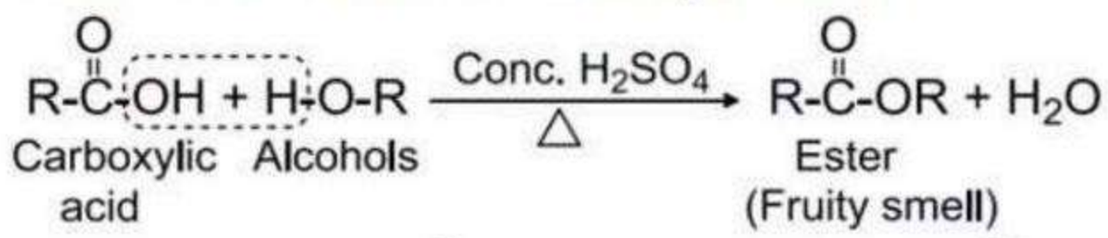
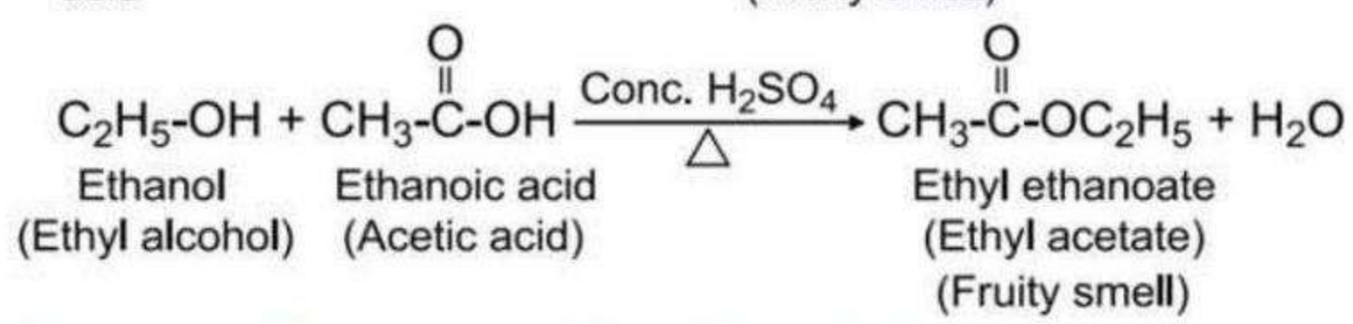
5. Esterification test or reaction for alcohols and carboxylic acids

When alcohol and carboxylic acid are heated with Conc. H_2SO_4 as catalyst then ester having pleasant fruity smells is formed. This is the laboratory test of alcohols as well as carboxylic acids.





6. Litmus test for carboxylic acids and phenols

Carboxylic acid and phenol both are acidic in nature and turn blue litmus paper into red.

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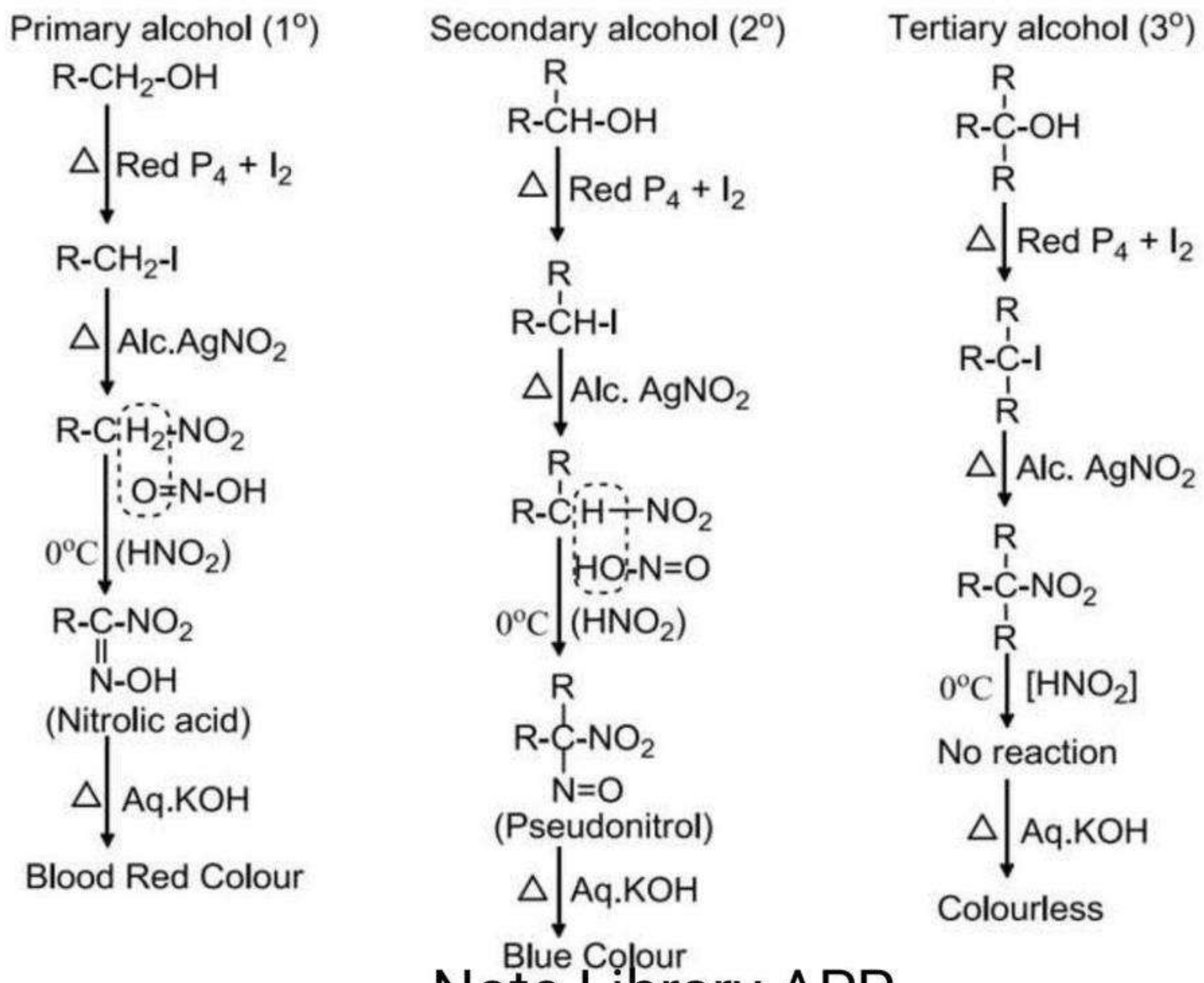

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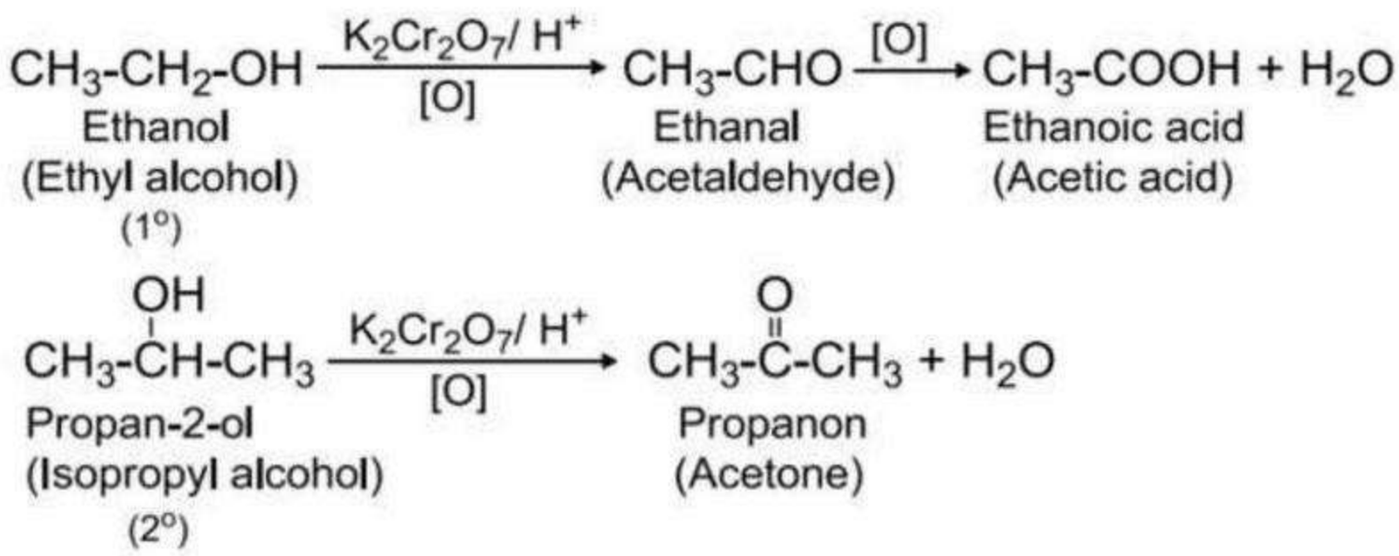
3. Oxidation test method for 1°, 2° & 3° alcohols

Primary, secondary and tertiary alcohols are distinguished by oxidation test method. In this method different alcohols are treated with mild oxidizing agents like acidified $KMnO_4$ or $K_2Cr_2O_7$ solution to form three different products.

- Primary alcohol gives aldehyde followed by carboxylic acid.
- Secondary alcohol gives ketone.
- Tertiary alcohol does not oxidize.



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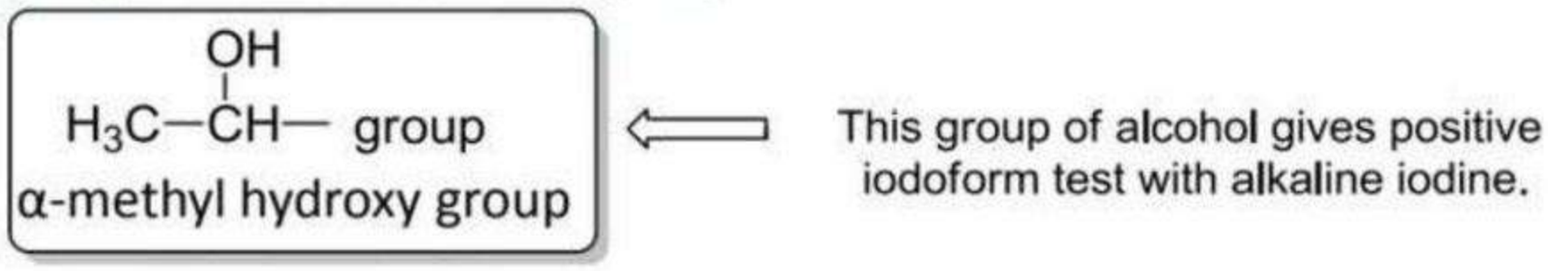
Some Important & Special Test Reaction

1. Iodoform test or reaction for alcohols

When alcohol is warmed with iodine in presence of aqueous NaOH or Na_2CO_3 solution at about 60-70°C then yellow crystalline precipitate of iodoform having hospital smell is formed. This is the test reaction of alcohol having α-methyl hydroxyl group.



Ethanol (Ethyl alcohol) is only one primary alcohol that gives positive iodoform test. Secondary alcohol having 2-hydroxy group forms yellow crystals of iodoform with NaOH and I_2. (Tertiary alcohol does not give iodoform test due to absence of oxidizable hydrogen)

$$\underset{\text{Ethanol (1°) (Ethyl alcohol)}}{CH_3CH_2OH} \xrightarrow[\triangle]{6NaOH + 4I_2} \underset{\text{Iodoform (Yellow crystal)}}{CHI_3\downarrow} + HCOONa + 5NaI + 5H_2O$$

$$\underset{\text{Propan-2-ol (2°) (Isopropyl alcohol)}}{CH_3\text{-}\overset{OH}{\overset{|}{C}}H\text{-}CH_3} \xrightarrow[\triangle]{4NaOH + 3I_2} \underset{\text{Iodoform (Yellow crystal)}}{CHI_3\downarrow} + CH_3COONa + 3NaI + 3H_2O$$

Therefore, this test is used to identify primary and secondary alcohols having α-methyl hydroxy group.

2. Victor-Meyer's test for the distinction of 1°, 2° & 3° alcohols

Note Library APP

Primary, secondary and tertiary alcohols are distinguished by Victor-Meyer's method. In this method different alcohols are treated with different chemical reagent to form three different products with respective colors.

- Primary alcohol gives blood red color.
- Secondary alcohol gives blue color.
- Tertiary alcohol does not give any specific color.

The chemical reagents used for this process are RANA

(R=Red $P_4 + I_2$) (A= Alc. $AgNO_2$) (N=Nitrous acid/HNO_2) (A=Aq. Alkali/KOH)

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